

AUTOMATION CASE STUDIES · 2024 TO 2026

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Automation and AI engineer. Eight systems I designed and built that run in production for US businesses today.

25

AUTOMATED
PROCESSES
RUNNING IN
PRODUCTION

300–400

TASKS COMPLETED
A DAY, NO PERSON
INVOLVED

0%

OF THOSE TASKS
FAILED

40s

TO ONBOARD A NEW
CLIENT, DOWN
FROM 40 MIN



Measured on a production environment I built for a client. Running today.

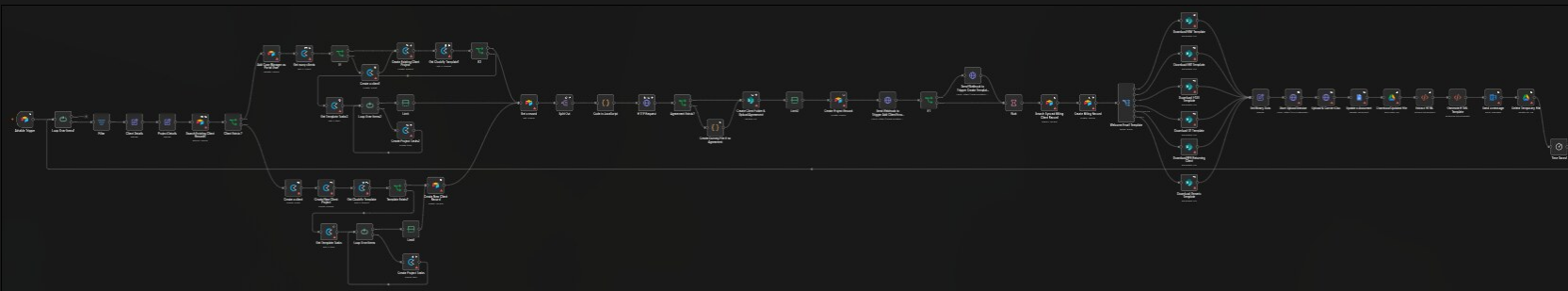
Client onboarding, 40 minutes to 40 seconds

PROBLEM Every new client meant 25 to 40 minutes of retyping the same details into the CRM, billing, time tracking and a welcome packet, by hand, by whoever was free.

BUILD One Airtable trigger. Dedupe against existing clients, create the client, project and billing records, set up the Clockify project, pick the document template by case type (five variants), generate it, upload to SharePoint through a chunked upload session, send the welcome email. 60 nodes, 33 with retry or error handling.

RESULT **About 40 seconds per client, unattended, since mid-2024.**

n8n · Airtable · Clockify · Google Docs · SharePoint · Outlook

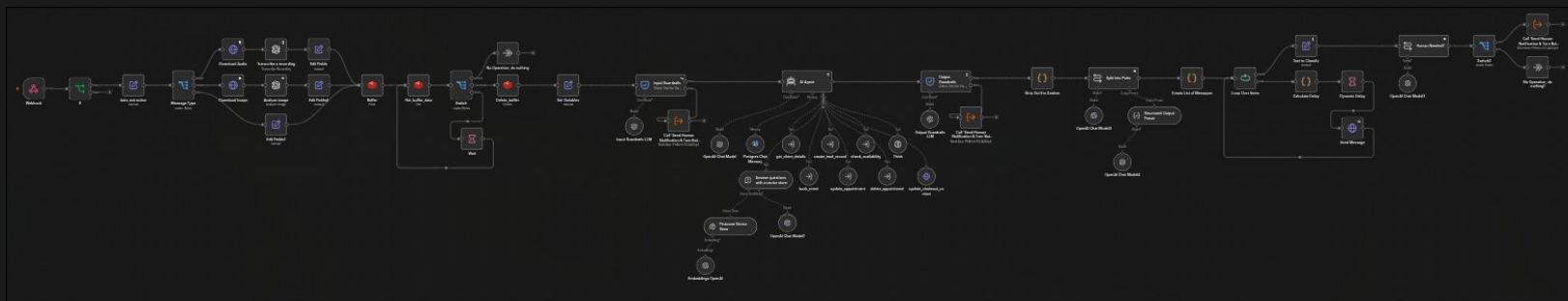


Workflow canvas. Client data removed.

Multi-channel support agent with guardrails

- PROBLEM** Enquiries arrived on WhatsApp, Instagram and Messenger as text, voice notes and screenshots. Answering them meant a person watching several inboxes.
- BUILD** One agent behind Chatwoot handles every channel. Audio is transcribed and images described before the model sees them, so all three input types become one conversation, and Redis buffers rapid-fire messages into a single turn. Input and output guardrails wrap the agent, it answers from a vector store, and it calls its own tools to look up a client, check availability and book, update or cancel an appointment. A classifier hands the thread to a person when one is needed.
- RESULT** **Text, voice and image handled on three channels, with a human handoff that triggers on intent rather than on a keyword.**

n8n · OpenAI · Chatwoot · Redis · Postgres · Airtable



Workflow canvas. Client data removed.

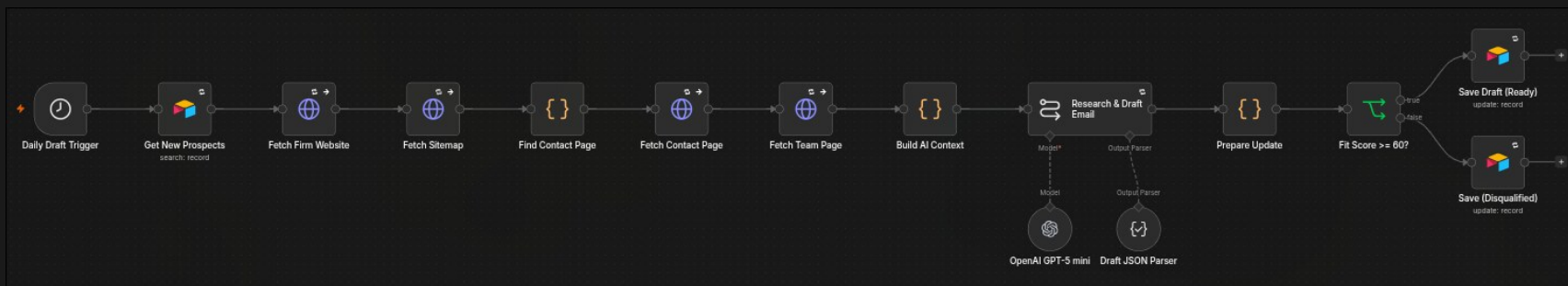
Outbound system, scrape to reply

PROBLEM Prospecting law firms by hand: find them, research them, write to them, follow up, track replies. Nobody did it consistently.

BUILD Six workflows. Weekly Google Maps scrape and dedupe. A research pass that fetches the firm's site, finds the contact and team pages with a sitemap fallback, then one LLM call scores fit and drafts a personalised email against a JSON schema. Deterministic fallbacks whenever the model's choice fails validation. Business-hours sequencing, delivery-event webhooks, and AI reply classification that routes interested leads and suppresses unsubscribes.

RESULT **Runs itself. The email picker never trusts the model alone.**

n8n · Apify · GPT-5 mini · Airtable · Resend



Workflow canvas. Client data removed.

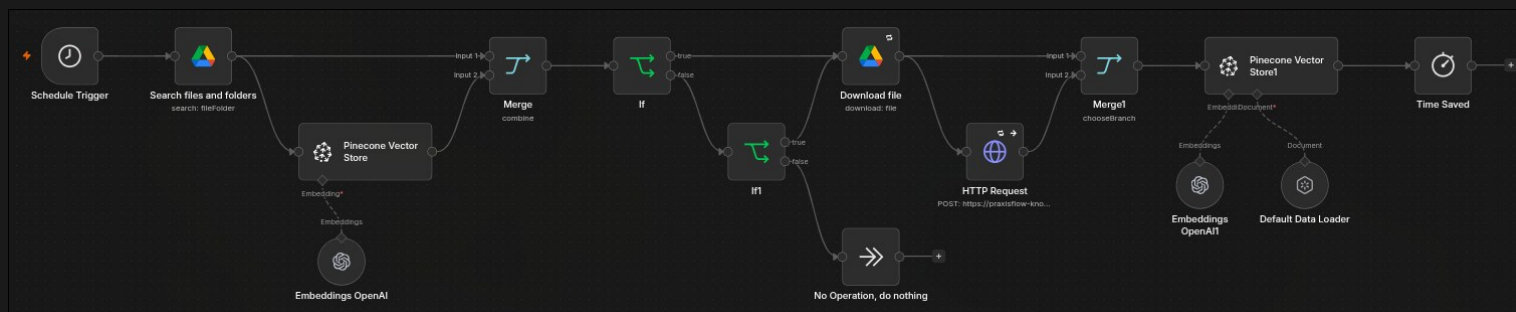
Documents that answer questions, kept current

PROBLEM Reference documents sat in a shared Drive folder. Finding an answer meant knowing which file it was in and reading it.

BUILD A scheduled pass over the folder. Each document is checked against what the vector store already holds, so only new and changed files are processed: text is extracted, split into overlapping passages, embedded and upserted, which replaces an edited file's old vectors rather than duplicating them. An assistant then answers from the retrieved passages and cites them.

RESULT **Drop a file in a folder and it is searchable within minutes. Edit it and the answers change with it.**

n8n · Google Drive · OpenAI embeddings · Pinecone



Workflow canvas. Client data removed.

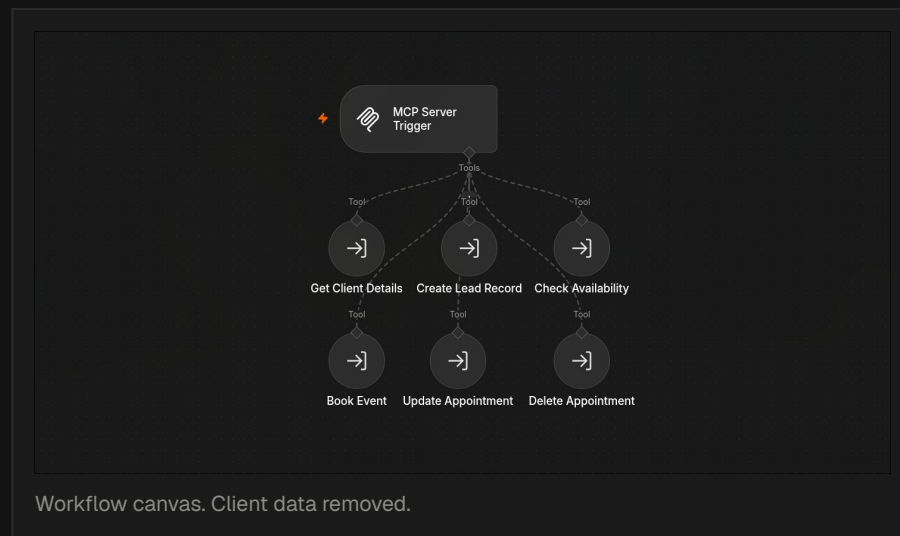
Voice agent with seven tools over MCP

PROBLEM A phone assistant that can only talk is a novelty. One that can look up a client, check availability and book is useful.

BUILD A Vapi voice agent connected to an MCP server exposing seven n8n workflows as callable tools: create lead, get client details, check availability, book, update, delete and look up appointments. The agent decides which to call; each tool is a small workflow with its own validation.

RESULT **Appointments booked and records created from a phone call, with every action logged.**

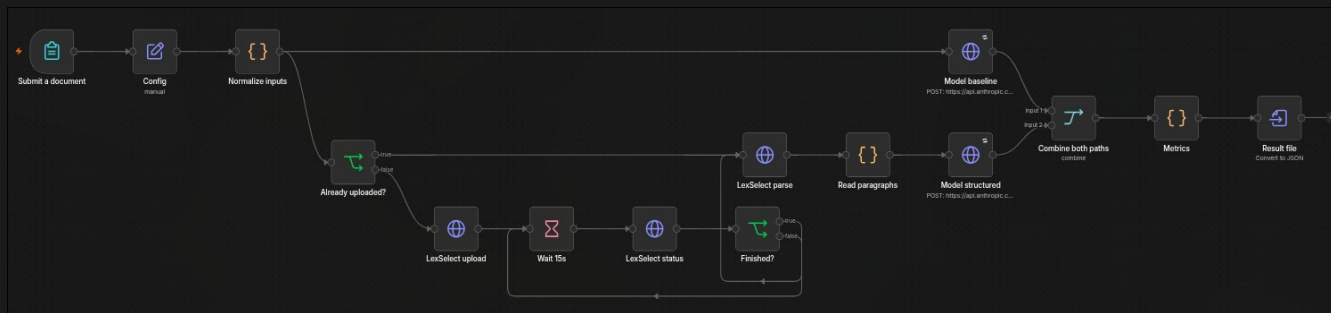
n8n · Vapi · MCP · Airtable · Google Calendar



Measuring a legal-AI parser instead of trusting it

- PROBLEM** A client needed to know whether a document-parsing API was worth adopting. The vendor's numbers were the vendor's.
- BUILD** An n8n workflow that runs one document down two paths, plain extracted text and the vendor's structured output, through the same model with the same prompt, and a verification step that checks every claim against the source PDF. Six documents, two runs each.
- RESULT** **51 to 93% fewer input tokens, 24 of 24 citations correct, one reproducible defect the vendor confirmed and adopted into their testing. The numbers are now in their public benchmark.**

n8n · Claude API · pdftotext



Workflow canvas. Client data removed.

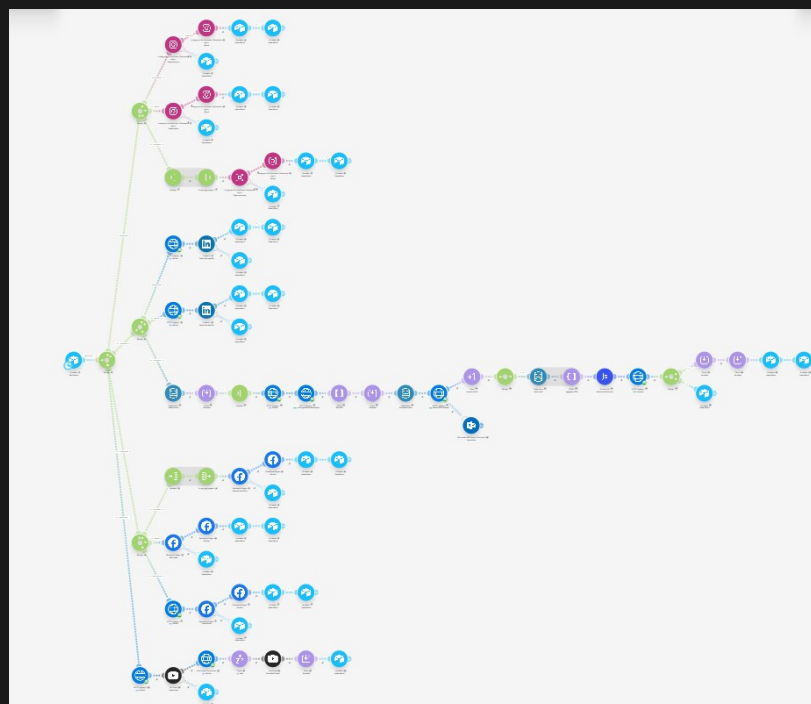
Scheduled publishing across five networks

PROBLEM Social posts were written in advance and then published by hand, one network at a time, each with its own formatting rules.

BUILD A Make scenario on a schedule. It reads the CRM for posts whose status and publish time are due, then routes each one to the right network and format: Instagram, Facebook, LinkedIn text and LinkedIn carousels, YouTube and a blog. LinkedIn was the hard part, both its text formatting and its carousel upload, and took working through their API documentation. The CRM table and its layout were built alongside the scenario.

RESULT **Posts written once, scheduled once, published everywhere on time.**

Make · Airtable · LinkedIn API · Meta Graph API



Make scenario. The branch per network and format.

Knowing it works, not hoping it does

PROBLEM Automation that fails quietly is worse than no automation. Somebody has to notice before the client does.

BUILD A Grafana dashboard per client instance, querying the n8n Postgres database directly in SQL. Instance health, active workflows, executions and failures in the last day, success rate, average execution time, database size, plus event-loop lag, memory and CPU for the host, and a per-workflow breakdown of runs and runtime.

RESULT **Failure rate is a number on a screen, not a surprise. Every figure on the cover of this deck comes from here.**

Grafana · PostgreSQL · Prometheus · self-hosted n8n



Grafana, one instance, last 24 hours. Queries the n8n database in SQL.

GET IN TOUCH

Let's build something that keeps running.

Looking for a role where I own a company's automation and AI layer end to end, and get measured on whether it keeps working. Remote, US hours, English and Spanish.

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Templates github.com/janr0599/n8n-templates